

Operations Research 2014/2015

Date: 11/02/2015

Exam – Second Call

Duration: 2 hours

Note: Present **all the calculations** that you perform and appropriately **justify** your answers.

1. Consider the following problem:

“The manager of a new bakery company, FAZPÃO, is designing a scheme of work shifts for his employees. The company must work continuously and, to this end, the day is divided into six shifts of eight hours each: 00:00-08:00, 04:00-12:00, 08:00-16:00, 12:00-20:00, 16:00-00:00, 20:00-04:00. The manager estimates that the minimum number of workers required during the day is the one presented in the following table:



Hours	Minimum Requirements
00:00 - 04:00	5
04:00 - 08:00	7
08:00 - 12:00	10
12:00 - 16:00	8
16:00 - 20:00	11
20:00 - 00:00	4

The agreement with the Union, for shift work, stipulates that each worker must be allocated to one of the shifts and must remain in the same, throughout his stay in the company.

In this scenario, the manager intends to hire the minimum number of workers which guarantees the fulfilment of all the above requirements.”

To help the manager of FAZPÃO, formulate the problem in terms of a linear programming model, indicating the meaning of the decision variables and objective function.

2. Consider the following linear programming problem:

$$\text{Minimize } z = x_1 + 2x_2 + 4x_3$$

subject to.

$$x_1 + 3x_2 \leq 6$$

$$x_1 + 3x_3 \geq 3$$

$$x_2 + x_3 \leq 6$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

- Solve the problem by the Simplex method with the “Big M” technique;
- Formulate the dual problem that corresponds to the one presented above;
- Without solving the dual problem, present its optimal solution as well as its objective function value.

Scores: 1 – 3,5 points 2 – 5,5 points 3 – 5,5 points 4 – 5,5 points

Departamento de Engenharia Informática e de Sistemas

3. Consider now the following linear programming problem:

Minimize $z = x_1 + x_2$
subject to
 $-x_1 + x_2 \geq 1$
 $2x_1 + 2x_2 \leq 6$
 $2x_1 - 3x_2 \geq 6$
 $x_1 \geq 0, x_2 \geq 0$

- a) Solve it by the graphical method;
- b) Solve it now by using the dual Simplex method;
- c) Please refer, concisely, the relationship between the complementary basic solutions of the primal and dual problems stated by the complementary slackness theorem.

4. Suppose that FAZPÃO company has expanded and now has two bakeries **P1** and **P2**, for sale to the public, from where bread is provided at low cost for three school canteens in the area, **C1**, **C2** and **C3**. It is known that, daily, **P1** and **P2** only have **100** and **70** units of bread, respectively, to sell at a special price. Moreover, the daily needs of **C1**, **C2** and **C3** are of **60**, **40** and **80** units of bread, respectively. The unitary transportation costs (in monetary units – MU) are given in the following table:



	C1	C2	C3
P1	7	4	5
P2	3	1	6

The company wants to know which is the scheme that it should use to daily transport the bread from the bakeries to the three canteens, in order to minimize the total transportation cost (that will be of its fully responsibility).

- a) Determine an initial basic feasible solution for the problem using the Vogel's approximation method (based on penalties calculation);
- b) Starting from the solution obtained in a), solve the problem by the transportation method;
- c) Interpreting the solution achieved in b), indicate how many bread units will be daily delivered to each of the canteens and also indicate the bakeries that will provide these bread units.